TESLA VALVE

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1. Introduction

The use of microvalve to move small quantities of fluid has great importance in the field of medicine, such as for example in drug delivery and in cardiology [1]. In these applications, the ability to control and move milli and even micro liters of fluid gains great importance, together with the ability to predict the behavior of the flow inside the valves.
Despite the facility of fabrication of many valves, their operation is characterized by the equations of moving fluids, in the form of the Navier-Stokes equation, whose solution is still an open problem in the field of mathematics. To predict the behavior of a moving fluid, a Computational Fluid Dynamics Simulation must be performed, whose aim is to solve numerically the equation of fluid.
In this paper we will analyse and discuss the diodicity of a Tesla valve with a CFD simulation using the open source computing platform FEniCS.

2. Tesla valve

One of the first design of micro valves is due to Nikola Tesla who patented a passive valve that allows flow preferentially in one direction, while it prevents flow in the other direction. The basic design of this valve consists of channel that is splitted in a Y junction that re-enters in a T channel This valve has many advantages since it is easy to design, easy to fabricate and it can be scaled with ease.

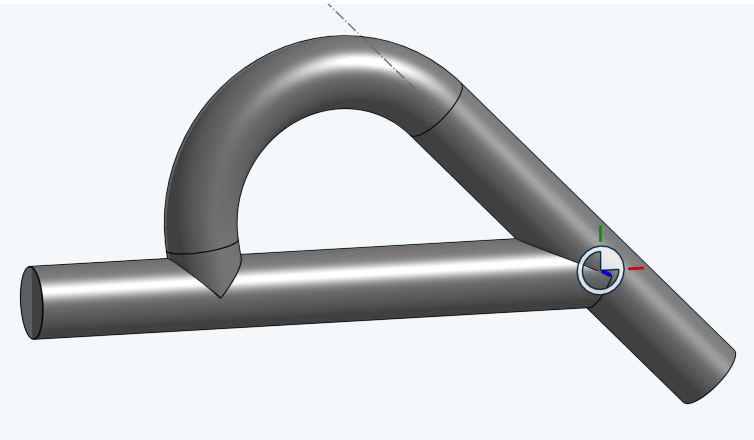


Figure 1. Tesla valve design used for this simulation.

Such a valve, named after its inventor, is structured to have a higher pressure drop for the flow in one direction (reverse) than the other (forward). This difference in flow resistance causes a net directional flow rate in the forward direction in oscillating flows. The efficiency is often expressed in diodicity D_i, defined as the ratio of pressure drops for identical flow rates

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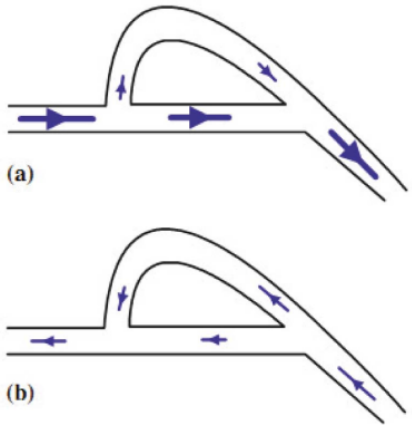


Figure 2. Flow in a tesla valve in forward (a) and reverse (b) direction [2]

$$D_i = \left(\frac{\Delta p_r}{\Delta p_f}\right)_Q \tag{1}$$

where Δp_r is the reverse flow pressure drop and Δp_f the forward flow pressure drop for flow rate.

Since the diodicity of such valve depends on the pressure drop between direct flow versus the reverse flow, higher diodicity values may be obtained by changing geometrical parameters of this type of valve, such as

- angle between channel and T junction
- angle between channel and Y junction
- number of branches of the valve

ADD COMMENT ABOUT SIMULATION.

3. Problem definition

The flow equations are the Navier Stokes equations for incompressible flow, which in the weak form can be stated as

$$\dot{u} + \nabla u \cdot u - \nu \Delta u + \nabla p = f \tag{2}$$

$$\nabla \cdot u = 0 \tag{3}$$

where u is the unknown velocity, p is the pressure field, ν is the given kinematic viscosity and f a given source.

paired with Dirichlet and Neumann boundary conditions that we will describe later.

The Navier Stokes equations can be rewritten in the finite element weak form as

$$F = \left(\frac{u - u_0}{dt}, v\right) + (\nabla u_m \cdot u_m, v) + \nu\,(\nabla u_m, \nabla v) + (\nabla p, v) + (\nabla \cdot u_m, q) - (f, v) \tag{4}$$

where $u_m = \theta u + (1 - \theta) u_0$ with $\theta = 0.5$ in the Crank-Nicholson scheme.

3.1. **Stabilization method.** It is well known that the numerical solution of Navier-Stokes equations using Finite Element Methods suffers from two sources of numerical instabilities [3] [4]

- using wrong combination of velocity and pressure finite element spaces (that do not satisfy the inf-sup condition) generate numerical oscillations in the pressure field (spurious pressure)
- when the problem is convection-dominated ($Re >> 1$), numerical oscllations in the velocity field occurs (called spurious velocity).

These sources of instabilities may be controlled in two ways

- using appropriate combination of velocity and pressure finite element spaces
- adding stabilization terms to the weak formulation of the problem

The first item can be addressed by using a P_k-$P_k + 1$ combination of spaces for the unknown fields velocity and pressure. Such mixed formulation is expressed in FEniCS in the following

```
Ve = VectorElement("CG", mesh.ufl_cell(), 2)
Qe = FiniteElement("CG", mesh.ufl_cell(), 1)
W = FunctionSpace(mesh, Ve * Qe)

# Define test functions
w = Function(W)
u, p = split(w)
(v, q) = TestFunctions(W)
w0 = Function(W)
u0, _ = split(w0)
```

in which the Vector Element space $Q_e = P_1$ for the pressure and the space $V_e = P_2$ for the velocity and mixed in the space $W = V_e Q_e = P_k P_{k+1} = P_1 P_2$.

The second way to reduce numerical instabilities is to add to the weak formulation a stabilization term as expressed in [3], [4] and [5]. In our formulation the stabilization term is the following

$$F_{stabilization} = \tau \left[(\nabla p + \nabla u_m \cdot u_m - f, \nabla q + \nabla v \cdot u_m) + (\nabla \cdot u_m, \nabla \cdot v) \right] \tag{5}$$

where τ is the stabilization parameter defined as

$$\tau = \left[\left(\frac{2}{dt} \right)^2 + \left(\frac{2\,\|u\|}{h} \right)^2 \left(\frac{4\nu}{h^2} \right)^2 \right]^{(-0.5)} \tag{6}$$

in which h is defined as the mean mesh cell diameter.

4. Geometry

5. Software

The CAD geometry has been designed using OnShape whose 3D step file has been imported into GmSH to define the mesh and surface boundaries. The calculations are performed using FEniCS open source computing platform and Paraview for the visualization of the results.

6. Results and discussion

References

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